DO NOW — RECALL

Lesson 099 · CISC and RISC: Instruction Set Design

Name: _____ Class: _____ Date: _____

1. What is ISA?

2. What does an opcode identify?

3. What is a load/store instruction?

4. Simplify $(A \wedge B) \vee (A \wedge \neg B)$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Instruction Set Architecture—programmer-visible instruction operations.

2. Model answer

Which operation the CPU performs.

3. Model answer

Transfers values between registers and memory.

4. Model answer

A∧(B∨¬B)=A.

By the end, you can…

• CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

• RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

• Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Vocabulary — know these terms

ISA — Programmer-visible machine instruction set.

CISC — Complex Instruction Set Computer.

RISC — Reduced Instruction Set Computer.

load/store — Explicit memory-register data transfer.

NEW LEARNING

Read, annotate and explain

Observe

CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

Apply

RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

Evaluate

Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Model a RISC load-add-store sequence for A5,B8 and result C.

Follow the reasoning

LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State two common RISC and two common CISC characteristics. [4]

Question 2

Trace the abstract RISC sequence above when A=6,B=9, give R1,R2,R3,C. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate statement 'CISC processors cannot use pipelining and RISC always uses fewer instructions'. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Modern CISC implementations cannot use pipeline stages.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.